Indian Economy

Agriculture

Exam: **UPSC CSE (Prelims)**

1. Basics of Agriculture

Tier: **tier3** • Weightage: **0.5** • Total Qs: **0** • PYQs: **7** • Year span: **1994–2020** • Model: **gpt-4.1** • Updated: **2026-01-07 13:53**

TopicDemandMap: Indian Economy > Agriculture > Basics of Agriculture

Explicit Subtopics Tested

1. Major Agricultural Statistics and Structure

- **Evidence:** [7461, 7457, 7452]
- **Description:**
- Questions test knowledge of India's cropped area, cropping patterns (e.g., dominance of cereals), average farm size, and employment share in agriculture.
- Also includes understanding of agricultural density concepts (e.g., people per unit arable land).
- **Type of Demand:**
- Factual recall (statistical facts, definitions)
- Conceptual understanding (interpretation of agricultural terms and structural features)
- Application (distinguishing between types of density, interpreting employment data)

2. Agricultural Innovations and Technological Change

- **Evidence:** [191, 7446]
- **Description:**
- Covers technological progress in agriculture (e.g., substitution of steel for wooden ploughs).
- Includes concepts like bio-fortification (breeding crops for higher nutritional value).
- **Type of Demand:**
- Conceptual clarity (types of technological progress)
- Definition-based (bio-fortification)
- Application (identifying examples of technological change)

3. Key Contributors and Historical Developments

- **Evidence:** [2245]
- **Description:**
- Focuses on major figures in agricultural history, such as Norman Borlaug and the Green Revolution.
- **Type of Demand:**
- Factual recall (names, contributions, origins)
- Contextual understanding (role in Indian agriculture)

4. Livestock and Allied Activities

- **Evidence:** [7288]
- **Description:**
- Tests knowledge of major Indian livestock breeds, specifically high milk-yielding goat breeds.
- **Type of Demand:**
- Factual recall (breed names and characteristics)
- Application (matching breeds to productivity traits)

Logical Study Sequence

1 **Livestock and Allied Activities** (7288)

- Foundation: Basic knowledge of agricultural components beyond crops.

1 **Major Agricultural Statistics and Structure** (7461, 7457, 7452)

- Builds on understanding of the sector's composition and key metrics.

1 **Key Contributors and Historical Developments** (2245)

- Contextualizes structural changes and innovations.

1 **Agricultural Innovations and Technological Change** (191, 7446)

- Advanced: Requires understanding of both historical context and conceptual frameworks.

Implicit Concepts and Prerequisites

- **Basic Agricultural Terminology:**
- Understanding of terms like cropping pattern, farm holding, arable land, agricultural density.
- **Indian Agricultural Context:**
- Awareness of the role of agriculture in employment and economy.
- **Technological Progress Types:**
- Distinction between labour-augmenting and capital-augmenting innovations.
- **Green Revolution:**
- General knowledge of its impact and key personalities.
- **Livestock Productivity Metrics:**
- Criteria for evaluating breeds (e.g., milk yield).

Adjacent Subtopics (Potential Future Demand)

- **Soil Types and Fertility Management:**
- Given the focus on cropping patterns and bio-fortification, questions on soil management are a logical extension.
- **Irrigation Methods and Water Management:**
- Closely linked to agricultural structure and technological change.
- **Agricultural Inputs (Seeds, Fertilizers, Pesticides):**
- Could be tested in relation to innovations and productivity.
- **Allied Sectors (Fisheries, Poultry):**
- Expansion from goats to other livestock or allied activities.
- **Agricultural Policy and Schemes:**
- Following from employment and structural statistics, policy interventions may be tested.

Notes

- The PYQ count is moderate (7 questions), allowing for some subdivision but not exhaustive mapping.
- Subtopics are derived strictly from the evidence provided; broader or more granular topics may exist but are not reflected in this dataset.
- Some subtopics (e.g., soil, irrigation, policy) are suggested as adjacent based on logical progression, not on current PYQ evidence.

2. Agriculture Inputs

Tier: **tier2** • Weightage: **4.0** • Total Qs: **0** • PYQs: **12** • Year span: **2001–2020** • Model: **gpt-4.1** • Updated: **2026-01-07 13:53**

TopicDemandMap: Indian Economy > Agriculture > Agriculture Inputs

Explicitly Tested Subtopics

1. Agricultural Price Support Mechanisms

- **Evidence:** 1383, 594, 610, 774
- **Description:**
- Understanding of Minimum Support Price (MSP), procurement prices, and their role in government procurement and price stabilization.
- Awareness of which crops are covered under MSP.
- Knowledge of how government policies (MSP, trading, stockpiling, subsidies) affect agricultural prices and economic cost calculations.
- **Type of Demand:**
- Factual recall of definitions and mechanisms (e.g., MSP, procurement price).
- Application of concepts to identify which crops are included/excluded.
- Analytical understanding of how policy instruments influence market prices and costs.

2. Agricultural Inputs: Fertilizers and Their Regulation

- **Evidence:** 1596, 351, 755
- **Description:**
- Knowledge of India's position in fertilizer production.
- Understanding of fertilizer types (e.g., urea), their production processes, and government interventions (e.g., neem-coating).
- Awareness of input sources for fertilizer manufacturing (e.g., ammonia from natural gas, sulphur from oil refineries).
- **Type of Demand:**
- Factual knowledge of production rankings and input sources.
- Conceptual understanding of government policies (e.g., price administration, neem-coating rationale).
- Application to current policy context.

3. Agricultural Inputs: Seeds and Seed Policies

- **Evidence:** 198
- **Description:**
- Understanding of the Seed Village Concept and its objectives in improving seed quality and availability.
- **Type of Demand:**
- Conceptual clarity on policy objectives and implementation mechanisms.

4. Agricultural Inputs: Edible Oils and Oilseeds

- **Evidence:** 1785, 511
- **Description:**
- Factors influencing India’s dependence on edible oil imports.
- Trends in domestic production vs. imports and related government policies (e.g., customs duties).
- **Type of Demand:**
- Analytical reasoning about structural issues in oilseed cultivation.
- Factual recall of recent trends and policy measures.

5. Regulation of Agricultural Markets and Commodities

- **Evidence:** 2605, 125
- **Description:**
- Knowledge of statutory regulation of agricultural markets (e.g., APMC Act, Essential Commodities Act).
- Understanding of government’s role in fixing prices and regulating essential commodities.
- **Type of Demand:**
- Factual recall of relevant legislations.
- Application to identify correct regulatory frameworks.

Logical Study Sequence

1. **Regulation of Agricultural Markets and Commodities** (2605, 125)
2. **Agricultural Price Support Mechanisms** (1383, 594, 610, 774)
3. **Agricultural Inputs: Fertilizers and Their Regulation** (1596, 351, 755)
4. **Agricultural Inputs: Seeds and Seed Policies** (198)
5. **Agricultural Inputs: Edible Oils and Oilseeds** (1785, 511)

Implicit Concepts / Prerequisites

- Basic understanding of government intervention in agricultural markets (price support, procurement, subsidies).
- Familiarity with key agricultural input categories: seeds, fertilizers, edible oils.
- Knowledge of major agricultural legislations and their purposes.
- Awareness of the structure and functioning of agricultural markets (APMCs, buffer stocks, PDS).
- Understanding of the economic rationale behind import dependence and input subsidies.

Adjacent Subtopics (Potential Future Demand)

- Agricultural credit and insurance mechanisms.
- Irrigation and water management as agricultural inputs.
- Farm mechanization and technology adoption.
- Organic and bio-fertilizer policies.
- Post-harvest management and storage infrastructure.
- Impact of WTO and international trade agreements on agricultural inputs.

Notes

- The PYQ set covers a broad but not exhaustive range of agricultural inputs, with emphasis on price mechanisms, fertilizers, seeds, and oilseeds.
- Some subtopics (e.g., pesticides, irrigation) are not directly tested in this set, indicating either a gap or a potential area for future questions.
- The number of PYQs allows for moderate subdivision, but some areas (e.g., seeds) are represented by only a single question, limiting deeper mapping.

3. Types of Farming and Cropping Patterns

Tier: **tier3** • Weightage: **0.5** • Total Qs: **0** • PYQs: **12** • Year span: **1997–2023** • Model: **gpt-4.1** • Updated: **2026-01-07 13:54**

TopicDemandMap: Indian Economy > Agriculture > Types of Farming and Cropping Patterns

Explicit Subtopics Tested (with Evidence)

1. Methods and Types of Cultivation

- **Evidence:** [7280, 7460, 7448, 7436]
- **Description:** Covers identification and characteristics of cultivation methods such as transplanting, shifting cultivation (including regional names like Jhum, Milpa, Ladang), and their geographic prevalence.
- **Demand Type:**
- Factual recall of cultivation methods and their features
- Recognition of regional and global terminology
- Understanding environmental and soil impacts of specific methods

2. Commercial and Subsistence Farming Types

- **Evidence:** [7453, 7440, 7451]
- **Description:** Tests knowledge of different farming types (commercial vs. subsistence), including plantation agriculture, dairy farming, livestock ranching, and their regional prominence.
- **Demand Type:**
- Classification of farming systems
- Ability to distinguish between commercial and subsistence practices
- Geographical association of farming types with Indian states

3. Cropping Patterns and Mixed Farming Practices

- **Evidence:** [7450, 7441]
- **Description:** Focuses on cropping patterns such as mixed cropping, intercropping, and the principles of organic farming.
- **Demand Type:**
- Conceptual understanding of crop arrangement and farming practices
- Knowledge of ecological and input-related aspects of farming systems

4. Irrigation Methods and Their Regional Patterns

- **Evidence:** [7455]
- **Description:** Examines the relationship between different irrigation methods (tank vs. well irrigation) and their geographic distribution.
- **Demand Type:**

- Analytical reasoning about physical geography and irrigation suitability
- Understanding of historical and environmental factors influencing irrigation choice

5. Collective and Cooperative Farming Models

- **Evidence:** [1047]
- **Description:** Tests the concept and operational features of collective farming models such as 'Small Farmer Large Field'.
- **Demand Type:**
- Conceptual clarity on cooperative farming arrangements
- Application of definitions to practical scenarios

6. Mechanization and Investment Patterns in Agriculture

- **Evidence:** [7459]
- **Description:** Assesses understanding of the drivers behind mechanization and investment trends in Indian agriculture.
- **Demand Type:**
- Application of economic reasoning to agricultural practices
- Linking labor market trends to technological adoption

Logical Study Sequence

1 **Methods and Types of Cultivation**

(Foundational: basic methods, terminology, and regional prevalence)

1 **Commercial and Subsistence Farming Types**

(Builds on understanding of cultivation methods to classify farming systems)

1 **Cropping Patterns and Mixed Farming Practices**

(Introduces complexity in crop arrangement and sustainable practices)

1 **Irrigation Methods and Their Regional Patterns**

(Adds environmental and infrastructural context)

1 **Mechanization and Investment Patterns in Agriculture**

(Economic and technological dimensions)

1 **Collective and Cooperative Farming Models**

(Advanced: organizational and policy aspects)

Implicit Concepts / Prerequisites

- **Basic agricultural geography:**

Knowledge of Indian states and their agro-climatic features.

- **Soil and climate requirements:**

Understanding how soil types and climate influence farming practices.

- **Economic principles:**

Labor markets, investment incentives, and productivity in agriculture.

- **Environmental impact:**

Effects of farming methods on soil fertility and sustainability.

- **Terminology:**

Familiarity with both Indian and international terms for farming systems.

Adjacent Subtopics (Potential Future Demand)

- **Agro-ecological zoning and crop suitability:**

How different zones determine cropping patterns.

- **Contract farming and value chains:**

Beyond collective models, the role of corporates and market linkages.

- **Technological interventions in cropping patterns:**

Use of biotechnology, precision farming, and their impact on traditional patterns.

- **Government schemes for sustainable cropping:**

Policy incentives for crop diversification and sustainable agriculture.

- **Impact of climate change on cropping patterns:**

Adaptive strategies and resilience in Indian agriculture.

Notes

- The PYQs provide a moderate spread across cultivation methods, farming types, cropping patterns, irrigation, mechanization, and collective models.
- Subtopic granularity is limited by the number and diversity of questions; deeper subdivisions (e.g., specific crop systems) are not evidenced.
- Some advanced or emerging areas (e.g., contract farming, climate adaptation) are not yet directly tested but are logical extensions.
- The map above strictly reflects only the patterns and concepts evidenced in the provided PYQs.

4. Major Crops, Agri Production and Agri Export

Tier: **tier2** • Weightage: **1.5** • Total Qs: **0** • PYQs: **30** • Year span: **1993–2023** • Model: **gpt-4.1** • Updated: **2026-01-07 13:54**

TopicDemandMap: Indian Economy > Agriculture > Major Crops, Agri Production and Agri Export

Explicit Subtopics Actually Tested

1. Agro-Climatic and Soil Requirements of Major Crops

- **Evidence:** 7286, 7283, 7281, 7278, 7444, 7439, 7449, 7289
- **Demand:**
- Identification of optimal temperature, rainfall, and soil conditions for crops like rice, wheat, cotton, sugarcane, and others.
- Understanding crop-specific climatic and soil suitability.
- Application of knowledge to match crops with their required environmental conditions.
- Recognition of crop-specific physiological responses (e.g., sucrose content in sugarcane).

2. Major Crop Types, Varieties, and Families

- **Evidence:** 7290, 7447, 7442, 7443
- **Demand:**
- Classification of crops by family or variety (e.g., rice, coffee, silk types, wool sources).

- Knowledge of important crop species and their distinguishing features.
- Ability to sequence or categorize varieties by production or quality.

3. Crop Distribution and Production Patterns in India

- **Evidence:** 7284, 7282, 7285, 2134, 2214, 7449, 7437
- **Demand:**
- Identification of major producing regions for specific crops.
- Interpretation of maps and spatial data for crop distribution.
- Sequencing crops by national production volume.
- Knowledge of regional and temporal patterns (e.g., Green Revolution states, cropping seasons).

4. Crop Pests, Diseases, and By-products

- **Evidence:** 7287, 1647
- **Demand:**
- Matching crops with major pests and diseases.
- Understanding of crop by-products and their uses (e.g., molasses, bagasse).
- Application of knowledge about crop processing and its economic implications.

5. Agricultural Exports and Imports: Trends and Commodities

- **Evidence:** 1492, 1598, 1759, 625, 698
- **Demand:**
- Identification of leading export and import commodities by value.
- Awareness of India's global ranking in specific agri-exports.
- Understanding trade patterns and their economic significance.

6. Cropping Intensity and Land Use Patterns

- **Evidence:** 7458, 2604
- **Demand:**
- Definitions and calculations related to cropping intensity.
- Trends in cultivated land area for major crops.
- Application of basic agricultural statistics.

7. Government Policies and Support for Agriculture

- **Evidence:** 1048, 2643

- **Demand:**
- Knowledge of government interventions (e.g., Minimum Support Price, Mega Food Parks).
- Awareness of policy objectives and their impact on production and processing.

Logical Study Sequence

1. **Agro-Climatic and Soil Requirements of Major Crops**
2. **Major Crop Types, Varieties, and Families**
3. **Crop Distribution and Production Patterns in India**
4. **Crop Pests, Diseases, and By-products**
5. **Cropping Intensity and Land Use Patterns**
6. **Government Policies and Support for Agriculture**
7. **Agricultural Exports and Imports: Trends and Commodities**

Implicit Concepts / Prerequisites

- Basic understanding of Indian geography (states, regions, climate zones)
- Familiarity with agricultural terms (Kharif/Rabi, cropping intensity, net sown area)
- Ability to interpret maps and tabular data
- General awareness of agricultural production statistics and trends
- Understanding of economic concepts like value addition, export-import balance

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Adjacent Subtopics (Potential Future Demand)

- Impact of climate change on crop patterns and yields
- Organic and sustainable farming practices for major crops
- Role of biotechnology and improved varieties in crop production
- Post-harvest management and value chain for agri-exports
- Regional disparities and challenges in agricultural productivity
- Detailed analysis of minor/underutilized crops and their export potential

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Notes

- The subtopics above are strictly derived from the provided PYQs; no external syllabus or textbook headings have been used.
- Some subtopics (e.g., pests, by-products) are less frequently tested but remain conceptually distinct.
- The map reflects a broad but recurring demand for factual, conceptual, and applied knowledge, especially regarding crop requirements, production patterns, and trade.
- If further granularity is required (e.g., for a specific crop), more PYQs or focused analysis would be necessary.

5. Government Schemes and Reforms

Tier: **tier2** • Weightage: **3.0** • Total Qs: **0** • PYQs: **11** • Year span: **2009–2020** • Model: **gpt-4.1** • Updated: **2026-01-07 13:53**

TopicDemandMap: Indian Economy > Agriculture > Government Schemes and Reforms

Explicit Subtopics Tested

1. Minimum Support Price (MSP) and Price Fixation Mechanisms

- **Evidence:** 2356, 7454, 780, 7438
- **Description:** Questions test knowledge of the concept of MSP, the process and criteria for its determination, the role of the Commission for Agricultural Costs and Prices (CACP), and the distinction between administered, market, and support prices.
- **Type of Demand:**
- Factual recall of institutions and their roles (CACP, MSP)
- Understanding of price determination criteria (cost of production, market trends, etc.)
- Application of concepts to identify correct/incorrect statements about MSP procurement and price levels

2. Agricultural Price and Procurement Policies for Specific Crops

- **Evidence:** 114
- **Description:** Focus on the Fair and Remunerative Price (FRP) for sugarcane and the approving authority.
- **Type of Demand:**

- Factual knowledge of crop-specific price policies
- Identification of responsible government bodies

3. National Schemes for Agricultural Marketing and Food Security

- **Evidence:** 2356, 459
- **Description:** Covers government initiatives like the National Food Security Mission and e-National Agriculture Market (e-NAM), including their objectives and advantages.
- **Type of Demand:**
- Understanding scheme objectives and implementing ministries
- Application to real-world benefits (e.g., market access, food security)

4. Food Safety and Regulatory Frameworks

- **Evidence:** 524
- **Description:** Tests awareness of major food safety legislation and regulatory authorities (FSSAI, Food Safety and Standards Act).
- **Type of Demand:**
- Factual recall of legislative changes and institutional responsibilities

5. Land Reforms and Agricultural Policy Strategies

- **Evidence:** 653, 7456
- **Description:** Questions on the aims and features of land reforms and post-independence strategies to boost foodgrain production.
- **Type of Demand:**
- Conceptual understanding of policy objectives and their implementation
- Ability to distinguish between different reform strategies

6. Public Investment and Infrastructure in Agriculture

- **Evidence:** 772
- **Description:** Identification of what constitutes public investment in agriculture, including infrastructure and institutional development.
- **Type of Demand:**
- Application of definitions to practical examples
- Distinguishing between direct investment and subsidies/waivers

7. Multi-purpose Irrigation and Development Projects

- **Evidence:** 7445
- **Description:** Knowledge of inter-state multi-purpose projects and their collaborative nature.
- **Type of Demand:**
- Factual recall of project partnerships and their geographic distribution

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Logical Study Sequence

1. Land Reforms and Agricultural Policy Strategies (653, 7456)
2. Multi-purpose Irrigation and Development Projects (7445)
3. Public Investment and Infrastructure in Agriculture (772)
4. Food Safety and Regulatory Frameworks (524)
5. National Schemes for Agricultural Marketing and Food Security (2356, 459)
6. Minimum Support Price (MSP) and Price Fixation Mechanisms (2356, 7454, 780, 7438)
7. Agricultural Price and Procurement Policies for Specific Crops (114)

Implicit Concepts / Prerequisites

- Basic understanding of agricultural economics (price mechanisms, market structures)
- Familiarity with government structure and roles of ministries/committees
- Knowledge of post-independence agricultural history and reforms
- Awareness of the difference between central and state government schemes
- Understanding of public investment vs. subsidies/transfer payments
- Concepts of food security and marketing infrastructure

Adjacent Subtopics (Potential Future Demand)

- Direct income support schemes for farmers (e.g., PM-KISAN)
- Crop insurance and risk mitigation schemes
- Agricultural credit and cooperative reforms
- Organic farming and sustainable agriculture initiatives
- Digital technology in agriculture (beyond e-NAM)

- Farmer Producer Organizations (FPOs) and collective marketing
- Implementation challenges of MSP and procurement

Notes

- The subtopic structure is based strictly on the evidence from the provided PYQs; no external syllabus headings have been introduced.
- Some subtopics (e.g., MSP) recur with multiple facets (criteria, process, application), indicating high demand.
- The number of PYQs allows for moderate subdivision, but some areas (e.g., public investment, food safety) have only single-question representation, limiting granularity.
- Adjacent subtopics are suggested based on logical extension from current patterns, not on external sources.